

Zaw Win Paing

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Portfolio: <https://barbatosv2.github.io> | **GitHub:** <https://github.com/BarbatosV2>

PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of experience developing and deploying AI, computer vision, robotics, and embedded systems solutions across agriculture, automation, and intelligent sensing applications.

Experienced in designing production ready machine learning pipelines, training deep learning models, deploying edge AI systems, and building autonomous robotic perception solutions using Python, PyTorch, TensorFlow, ROS, NVIDIA Jetson, AWS, and cloud-native technologies.

Strong background in computer vision, reinforcement learning, robotics simulation, embedded systems, and fullstack software development. Proven ability to deliver end to end AI solutions from data collection, data annotation and model development through deployment and optimization.

CORE COMPETENCIES

Machine Learning | Deep Learning | Computer Vision | Reinforcement Learning | Robotics | ROS | Embedded AI | AWS | MLOps | Edge Computing | Object Detection | Autonomous Systems | Sensor Integration | Model Deployment | Python | C++ | Docker | Kubernetes

EDUCATION

The University of Adelaide, Adelaide, Australia

Bachelor of Computer Science (AI)

July 2019 – June 2023

Eynesbury College, Adelaide, Australia

Diploma of Computing and Information Technology

October 2018 – June 2019

EXPERIENCE

AirLabOne – Machine Learning Engineer

September 2024 – Present

Develop AI-powered laboratory automation systems incorporating computer vision, audio intelligence, robotics, and cloud infrastructure.

Key Achievements

- Developed real-time object detection, audio classification and event recognition systems enabling automated monitoring and analysis of laboratory operations.
- Built machine learning pipelines for data labeling, training, evaluation, and deployment, improving model readiness for production environments.
- Implemented cloud-based data collection infrastructure using AWS Kinesis, Rekognition, and EC2 to support scalable AI model development.
- Applied reinforcement learning techniques within NVIDIA Omniverse and ROS/Gazebo simulations to explore autonomous machine behaviors.
- Integrated machine learning solutions with NVIDIA Jetson, Arduino, and ESP-based edge devices for real-world deployment.

Technologies

Python, PyTorch, TensorFlow, AWS, Rekognition, Kinesis, EC2, ROS, Gazebo, NVIDIA, Omniverse, Jetson, Arduino, ESP32, Computer Vision

Flux Robotics, Adelaide, Australia – Computer Vision and Machine Learning Engineer

November 2022 – September 2024

Developed perception systems and machine learning models for agricultural robotics applications.

Key Achievements

- Designed and trained computer vision models for object detection and classification in robotic environments.
- Improved robotic perception accuracy through model validation, performance analysis, and iterative optimization.
- Built simulation environments using ROS and Gazebo to accelerate testing and reduce physical hardware dependency.
- Developed camera calibration algorithms enabling accurate distance estimation and environmental measurements.
- Integrated laser-based sensing systems and embedded hardware for precision detection applications.
- Created scalable data collection, annotation, and preprocessing pipelines supporting machine learning development.

Technologies

Computer Vision, OpenCV, Python, ROS, Gazebo, Embedded Systems, CUDA, Machine Learning, Robotics

TECHNICAL SKILLS

AI / Machine Learning

PyTorch, TensorFlow, Scikit-Learn, Keras, OpenCV, Reinforcement Learning, Computer Vision

Robotics & Simulation

ROS, Gazebo, NVIDIA Omniverse, Autonomous Systems, Sensor Integration

Programming

Python, C++, Java, C#, JavaScript, TypeScript, SQL, Bash, Matlab, R

Cloud & DevOps

AWS, Azure, Google Cloud Platform, Docker, Kubernetes, Git

Databases & Analysis

PostgreSQL, MongoDB, SQLite, SAS

AWARDS

- **Ideation Project (December 2020):**
Project: Thate Pan Hub – Teaching children computer science without technology. Won the "Most Effective Idea" award.
- **VR Project (July 2020 – November 2020):**
Developed a bug-free VR game where players can spawn and build objects. Contributed to sound recording, 3D modeling, UI design, and programming using C#.
- **Tech eChallenge (July 2021 – November 2021):**
Designed an AI scheduler to assist with time management and rescheduling. Achieved 4th place in the competition.